INCH-POUND

A-A-55278
2 May 1995

COMMERCIAL ITEM DESCRIPTION

COVERALLS, ANTI-EXPOSURE

The General Services Administration has authorized the use of this commercial item description as a replacement for MIL-C-29109 for all Federal agencies.

1. SCOPE. This commercial item description covers the requirements for a hooded submarine deck exposure coverall constructed with a coated nylon fabric outershell, a plastic foam interlining, and a nylon fabric lining. This coverall is intended for use by Navy topside personnel to provide thermal protection in extreme wet/cold environments.

2. SALIENT CHARACTERISTICS. The fully lined coverall shall have a flexible foam interlining and two fleece-lined hand warmer chest pockets; two cargo pockets with flaps (hook and loop tape closures) on front of garment; pencil pocket (four slot) with flap (hook and loop tape closure), on upper left sleeve; retro-reflective tape positioned vertically at center of attached hood, down the front panels, and horizontally across the center back of the garment for improved night time visibility; and a braided cord and attachable cord lock inserted within a 1/2 inch channeled hem on hood to secure closure around the neck. The braided cord shall measure 28-1/2 inches after ends are knotted. The two chest pockets and two cargo pockets shall have drain holes in the lower corners, both sleeves shall have four drain holes located on the lining according to the marks on the patterns. All drain holes shall be reinforced with corrosion-resistant eyelets.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Defense Personnel Support Center, Clothing and Textiles Directorate, Attn: DPSC-FQSC, 2800 South 20th Street, Philadelphia, PA 19145-5099.

AMSC N/A FSC 8415

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The front closure shall have a slide fastener and storm flap (hook and loop tape closure) from the neck to crotch of garment. In addition, each sleeve opening shall have a slide fastener closure and each leg opening shall have a slide fastener and storm flap (hook and loop tape closure). The leg bottoms, sleeves, and waistband shall have hook and loop adjustment straps. All hook and loop tape fasteners and retro-reflective tape shall be set according to marks on the pattern. The chest and cargo pockets shall have vertical bartacks at the openings.

2.1 Materials.

2.1.1 Outershell. The base material shall be a nylon cloth made from first quality, continuous filament yarn, and shall conform to the physical requirements of Table I. The finished vulcanized cloth shall be coated on one side with approximately 4.0 ounces per square yard (135.6 g/m^2) of chloroprene rubber suitably compounded; plasticized; and pigmented in an international orange shade. No elastomer other than chloroprene shall be used. The finished coated cloth shall conform to the physical requirements of Table II.

TABLE I. Uncoated cloth physical requirements

Characteristics	Minimum	Maximum	Test method
Weave	2 up 1 down right hand twill		Visual 1/
Yarn			
Denier	210	-	2/
Filament	30	-	2/
Luster	Bright	-	2/
Weight, oz./yd.^{2}	3.4	3.6	ASTM D3776 option C 5/
Yarns per inch			
Warp	56	-	ASTM D3775 5/
Filling	56	-	ASTM D3775 5/
Breaking strength, lbs.			
Warp	225	-	ASTM-D-5034
Filling	210	-	ASTM-D-5034
Flex stiffness, in.-lbs. x 10^{-4}			
Warp	-	2.5	FED-STD-191-5206
Filling	-	1.8	FED-STD-191-5206
Nonfibrous material	-	1.0%	ASTM D629 (section 9)
Heat setting test			
Distortion or puckering	No appreciable chg.		Visual 3/
Dimensional change	-	2.0%	3/ 4/
Colorfastness to:			
Laundering	"Good"	-	AATCC 61 test 1A 6/
Wet-dry cleaning	"Good"	-	FED-STD-191-5622
Light	"Good"	-	AATCC 8
Crocking, Munsell value	-	8.5	AATCC 16 option A 7/

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TABLE I. Uncoated cloth physical requirements - Continued

Characteristics	Minimum	Maximum	Test method
Sulfur content	No more labile sulfur than shown by the standard sample		FED-STD-191-2020
pH	5.0	8.5	FED-STD-191-2811
Copper content	-	0.003 %	FED-STD-191-2053
Manganese content	-	0.0015 %	FED-STD-191-2053

1/ One determination per sample unit and the results reported as "pass" or "fail".

2/ A certificate of compliance will be acceptable for this requirement.

3/ The test specimen shall consist of a square of cloth at least 12 by 12 inches marked off with a square 10 by 10 inches. The test specimen shall be placed in an oven maintained at 280°F ±2° for a period of 2 hours; then the specimen shall be removed from the oven and conditioned under standard atmospheric conditions. The specimen shall then be measured for dimensional changes. The measurements shall be made at least 1 inch from the edge of the square at the center. The specimen shall then be visually compared with the original unheated fabric for distortion, puckering, or change in color. The change in color shall be evaluated in accordance with Method 5662 of FED-STD-191. One specimen shall be tested from each sample unit. The dimensional change of the sample unit shall be the average of the specimens tested from each direction, respectively, and shall be reported separately to the nearest 0.1 percent. Distortion and puckering shall be reported as "pass" or "fail".

NOTE: When cutting the samples, make certain the twill line goes from the lower left to the upper right in a 45 degree angle.

4/ One determination per sample unit and the dimensional change reported to the nearest 0.1 percent.

5/ Samples shall be conditioned as directed in "D1776 Practice for Conditioning Textiles for Testing" prior to testing.

6/ The specimens must be dried after each of the three laundering cycles. The soap solution shall contain 0.5 percent P-D-245, Detergent, Laundry, and Hand Dishwashing (Granular) Type II, in water hardness not over 50 parts per million. The specimen weight shall be 4.0 ± 0.5 grams and the rotating time shall be 30 minutes with 10 stainless steel balls in each cylinder.

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7/ When a color change equal to Step 4 on the Gray Scale is evident in either evaluation sample or the standard sample, exposure shall be continued for an additional number of hours equal to the number of hours at which the color change was apparent. Specimens which do not show a color change until 80 AFU's shall be returned to the machine for a total exposure period not to exceed 140 hours.

TABLE II. Coated cloth physical requirements

Characteristic	Minimum	Maximum	Test method
Weight, oz./yd.^{2}	7.0	9.0	ASTM D3776 opt.C 11/
Breaking strength, lbs.			
Warp	225	-	ASTM D5034
Filling	210	-	ASTM D5034
Tear strength, grams			
Warp	4500	-	ASTM D1424 1/
Filling	3400	-	ASTM D1424 1/
Hydrostatic resistance, lbs./in.^{2}			
Initial	100	-	ASTM D751 2/ 12/
After 5 launderings	80	-	FED-STD-191-5556 3/4/
		-	ASTM D751 2/ 12/
After 40 hours weatherometer	80	-	FED-STD-191-5804 4/5/
		-	ASTM D751 2/ 12/
Flex stiffness, in.-lbs. X 10^{-4} at 70°F			FED-STD-191-5206 6/7/
Warp	-	9.0	
Filling	-	7.0	
Flex stiffness, inch-lbs X 10^{-4} at -20°F			FED-STD-191-5206 6/8/
Warp	-	25.0	
Filling	-	25.0	
Resistance to heat at 160°F	No evidence of tackiness, exudation, or loss of flexibility		FED-STD-191-5850 9/
Adhesion of coating, lbs per 2 inch width	10	-	FED-STD-191-5970
Blocking, scale rating	-	No. 2	FED-STD-191-5872
Dusting powder			
Resistance to dry crocking "good"		-	AATCC 8 10/

1/ Standard instrument with one augmenting weight.

2/ The water pressure shall be applied only to the uncoated side of the test specimens.

3/ After subjection to laundering or weathering, but prior to hydrostatic testing, the test specimens shall be visually examined for any evidence of cracking, blistering, stiffening, or delamination of the coating from the base cloth. Any evidence of these conditions shall be cause for rejection of the lot.

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4/ The cotton laundering procedure shall be followed.

5/ The coated side of the test specimens shall face the light source.

6/ Ten specimens shall be tested and averaged, five with the coated side up and five with the coated side down.

7/ Prior to performing this test, the test specimens shall be conditioned for at least four hours at the specified temperature.

8/ Prior to performing this test, the test specimens shall be conditioned for one hour at the specified temperature.

9/ The test specimens shall be heated in the oven for 120 hours at 160°F and then visually examined.

10/ The test shall be conducted using the blue/black crock cloth.

11/ Sampl8s shall be conditioned as directed in "D1776 Practice for Conditioning Textiles for Testing" prior to testing.

12/ Procedure A -Pressure application by Mullen type Hydrostatic Tester, sections 34-37, Procedure 1 (except that the rate of displacement shall be as specified in section 34.2) shall be used.

2.1.2 Foam interlining. The foam interlining shall be unicellular closed cell foam, 3/16 inch in thickness, and shall be made from an expanded supersoft polyvinyl-chloride plastic material. The finished material shall comply with US Coast Guard Regulation CFR 164.015 for type A foam.

2.1.3 Lining. The lining material shall be nylon taffeta fully heat set at a minimum temperature of 400°F (204°C). The shade shall be black. The yarn shall be continuous filament, semi dull or equal for the warp and filling and shall conform to the physical requirements of Table III when tested as specified.

2.1.4 Fusible interlining The fusible interlining utilized to impart stiffness to flaps and closures shall be of a type suitable for the intended end use.

2.1.5 Fleece lining The fleece lining for the chest pockets shall be made of 100 percent polyester, 6.4 oz./yd.^{2}, 32 gauge knit fabric.

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TABLE III. Lining cloth physical requirements

Characteristic	Minimum	Maximum	Test method
Weave	Plain	-	Visual 1/
Yarn denier	-	70	2/
Weight, oz./yd^2	1.9	2.0	ASTM D3776 opt. C 8/
Yarns per inch			
Warp	106		ASTM D3775 8/
Filling	92		ASTM D3775 8/
Break strength, lbs.			
Warp	110	-	ASTM D5034
Filling	95	-	ASTM D5034
Tear strength, lbs.			
Warp	3.64	-	ASTM D1424 3/
Filling	2.97	-	ASTM D1424 3/
Flex stiffness, in.-lbs. x 10^{-4}			
Warp	-	1.5	FED-STD-191-5206
Filling	-	0.5	FED-STD-191-5206
Dynamic Absorption (max.)			
Initial	8% max for avg. of sample units and 10% max. for any individual sample		AATCC 70
After 3 laundering			AATCC 96 9/ 10/ 11/ option Ic AATCC 70 4/
Spray rating (3 samples)			
Initial	100,100,90	-	AATCC 22 12/
After 3 launderings	80,80,70	-	AATCC 96 option Ic 9/ 10/ 11/ AATCC 22 4/ 12/
Nonfibrous material	-	1.0%	FED-STD-191-2611
Heat setting test			
Distortion or puckering	No appreciable chg		5/ 6/
Dimensional change	-	2.0%	5/ 6/
Colorfastness to:			
Laundering	"good"	-	AATCC 61 test 1A 13/
Dry cleaning	"good"	-	FED-STD-191-5620
Perspiration	"good"	-	AATCC 15 14/
Light	"good"	-	AATCC 16 opt. A 7/ 5/
Crocking	3.0	-	AATCC 8
pH	5.0	8.5	FED-STD-191-2811

1/ One determination per sample unit and the results reported as "pass" or "fail".

2/ A certificate of compliance will be acceptable for the stated requirement.

3/ Standard instrument with one augmenting weight.

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4/ Specimens shall be subjected to 3 complete cycles (wash and dry) prior to determination of dynamic absorption and spray rating.

5/ One determination per sample unit and the dimensional change reported to the nearest 0.1 percent.

6/ The test specimen shall consist of a square of cloth at least 12 by 12 inches marked off with a square 10 by 10 inches. The test specimen shall be placed in an oven maintained at 280°F ±2° for a period of 2 hours; then the specimen shall be removed from the oven and conditioned under standard atmospheric conditions. The specimen shall then be measured for dimensional changes. The measurements shall be made at least 1 inch from the edge of the square at the center. The specimen shall then be visually compared with the original unheated fabric for distortion, puckering, or change in color. The change in color shall be evaluated in accordance with Method 5662 of FED-STD-191. One specimen shall be tested from each sample unit. The dimensional change of the sample unit shall be the average of the specimens tested from each direction, respectively, and shall be reported separately to the nearest 0.1 percent. Distortion and puckering shall be reported as "pass" or "fail".

7/ Total exposure time shall be 20 hours.

8/ Samples shall be conditioned as directed in "D1776 Practice for Conditioning Textiles for Testing" prior to testing.

9/ Sample size shall be 22 by 22 inches with 18 inch benchmarks, 6 inches apart.

10/ Use Drying Procedure (D) or samples may be dried using a thermostatically controlled circulating oven at 221° - 230°F until dried. The sample shall be allowed to cool for 5 minutes, sprinkled with water to permit damp pressing, and allowed to stand in this condition for 5 minutes.

11/ The dampened specimens shall be pressed with a flatbed press as described in AATCC 96 Section 5.2 or a hand iron as described in AATCC 96 Section 5.10 at a temperature of 275° to 300° F. Section 7.4 of AATCC 96 shall not apply.

12/ Each sample shall be 7 inches by 7 inches.

13/ The specimen must be dried after each of the three laundering cycles. The soap solution shall contain 0.5 percent by weight, P-D-245, Detergent, Laundry, and Hand Dishwashing (Granular) Type II, in water hardness not over 50 parts per million. The weight of the specimen shall be 4.0 ±0.5 grams and the rotating time shall be 30 minutes with ten stainless steel balls in each cylinder.

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14/ The test for colorfastness for acid and alkaline solution shall apply.

15/ When a color change equal to Step 4 on the Gray Scale is evident in either evaluation sample or the standard sample, exposure shall be continued for an additional number of hours equal to the number of hours at which the color change was apparent. Specimens which do not show a color change until 80 AFU's shall be returned to the machine for a total exposure period not to exceed 140 hours.

2.1.6 Leg bottom mesh. The knit mesh material used for drainage at the bottom of the leg opening shall be Apex Mills Corporation Style D51 or equal. The knit mesh shall finish approximately 2 to 3 inches in height around the circumference of the leg bottom.

2.1.7 Draw cord. The drawcord with tip for securing the hood around the neck, shall be made of black polypropylene material, 1/8 inch in width, conforming to W.M. Jetty style #48 or equal.

2.1 8 Cord lock. The plastic cord lock to be used on the draw cord shall be Waterbury Buckle Co. #304-000 or equal.

2.1.9 Retro-reflective tape. The retro-reflective tape used on the hood, front panels and back shall be 3M "scotchlite" #8930 or equal.

2.1.10 Thread. The thread for seaming and stitching the coveralls shall be nylon conforming to Type II, Class A, size E of A-A-55216. The color of the thread shall approximate the shade of the outershell and the lining materials respectively.

2.1.11 Slide fasteners. The slide fasteners shall conform to the requirements of V-F-106, size MHS (individual element molded plastic with a minimum crosswise breaking strength of 160 lbs.) black in color and equipped with plastic sliders. The thong shall be 3/8 inch wide by 2 inches belt looped outershell material. Slide fastener lengths are 7 inches for sleeve openings and 14 inches for leg openings. The front slide fastener is graded and set according to the finished neck seam and the step at the crotch.

2.1.12 Front closure. The slide fastener for the front closure shall conform to Type I, Style 15 of V-F-106.

2.1.13 Sleeve openings. The slide fasteners for the sleeve opening shall conform to Type I, Style 3 of V-F-106.

2.1.14 Leg openings The slide fasteners for the leg openings shall conform to Type I, Style 3 of V-F-106.

2.1.15 Eyelets. The corrosion-resistant eyelets for the drainholes shall have an inside diameter of 7/32 inch and a black chemical finish.

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2.1.16 Hook and loop fastener tape. The matching hook and loop fastener tape shall conform to type II, class 3, of MIL-F-21840. The color shall be black.

2.2 Label/marking. The label shall be centered on the inside of the back with the top edge of the label parallel to and 1-1/2 to 2 inches below top raw edge of the back. The lining material, only, shall be stitched on all four sides of label, 1/8 inch from the edge. A commercial-type label is acceptable, provided that the following information is included on the commercial label.

COVERALLS, ANTI-EXPOSURE
FIBER CONTENT: 100% NYLON
SIZE: Small (Example)
STOCK NUMBER: 8415-00-000-0000 (Example)
CONTRACT NUMBER: DLA-NU-00000-0 (Example)
NAME OF CONTRACTOR:
LAUNDER BY HAND IN WARM WATER WITH MILD SOAP AND HANG TO DRY.

2.2.1 Label/tag. Each item shall be individually bar-coded with a paper tag for personal clothing items. The paper used for the tags shall be bleached sulfate having a basis weight of 100 pounds. The paper used for the tags shall have a smooth finish to accept thermal transfer and direct printing. The tag shall have a hole and shall be attached to each item by a fastener, clearly legible and readable by a scanner. The bar coding element shall be a 13 digit national stock number (NSN). The bar code type shall be a medium to high code density and shall be located so that it is completely visible on the item when it is folded and/or packaged as specified and so that it causes no damage to the item.

2.3 Seams and stitching. The coveralls shall be constructed using the 301 stitch type. The stitches per inch for all operations shall be 8 to 10. All hook and loop tape and retro-reflective tape shall be stitched 1/8 to 3/16 inch from the edge. The bottom and sides of pocket flaps shall be double stitched 3/16 to 1/4 gage, 1/16 to 1/8 inch from the edge. The top of the pocket flaps shall be stitched down 3/8 from the edge to bury raw edged within the seam. All pockets shall be stitched 1/16 to 1/8 inches from the edge. The hood back seams shall be turned toward the center and top stitched 1/16 from the edge.

2.4 Measurements. The finished dimensions of the coveralls shall be as indicated in Table V. All measurements are expressed in inches.

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TABLE V. Finished measurements

Size	Chest A	Leg inseam B	Length C	Sleeve Inseam D
Small (36-38)	45	29	62	22
Medium (40-42)	49	29-1/2	63-3/4	22-1/2
Large (44-46)	53	30	65-1/2	23
X-Large (48-50)	57	30-1/2	67-1/4	23-1/2
XX-Large (50-52)	59	31	69	24
Tolerance	± 1	± 1/2	± 3/4	± 1/2

Note: Refer to Figure 1 for A, B, C, and D.

A. Twice the measurement taken across the front at base of armholes, from folded edge to folded edge.

B. Measurement shall be taken from the junction of crotch and seat seam along inseam to bottom of leg.

C. Measurement shall be taken from the intersection of shoulder and hood seams to bottom of hemmed leg.

D. Measurement shall be taken from base of armhole along sleeve inseam to bottom of sleeve.

2.5 Patterns. The Government shall furnish a single size pattern. The contractor is responsible for grading patterns for additional sizes. The Government patterns shall be used to make the contractor's working patterns. The Government patterns shall not be altered in any way. The working patterns may be modified to accommodate automation, however, if the working patterns are modified that design and measurements must be maintained as intend by the Government patterns provided they are approved for use by the Government and conform to the cited design and finished measurements.

2.6 Equal item. Prior to the use of an "or equal" item the contractor shall submit the item with supporting data to the contracting officer for subsequent approval or disapproval by the responsible Military Agency.

3. REGULATORY REQUIREMENTS. The offeror/contractor is encouraged to use recovered materials to the maximum extent practicable, in accordance with paragraph 23.403 of the Federal Acquisition Regulation (FAR).

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4. QUALITY ASSURANCE PROVISIONS

4.1 Certification. The contractor shall certify and maintain substantiating evidence that the product offered meets the salient characteristics of this commercial item description; and that the product conforms to the producer's own drawings, specifications, standards, and quality assurance practices; and is the same product offered for sale in the commercial marketplace. The Government reserves the right to require proof of such conformance prior to first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

4.2 Market acceptability criteria. The item offered, or a generic equivalent, must have been sold to the commercial market or to the Government for at least 2 years.

4.3 Visual examination. Each coverall shall be examined for the defects listed below.

Material defects	Any ply having a hole, cut, tear, smash, burn or needle chew; any ply having a darn or mend; any exposed drill holes (in any ply); one or more rows of stitching omitted in the stitching
Cleanness	Any spot or stain on outside; thread ends not trimmed or loose threads not removed
Component and assembly	Any component part missing, tight, twisted, distorted, puckered, or pleated; pockets horizontally misaligned, slide fastener operation not smooth
Seams and stitching	Any seams twisted, loose, puckered, staggered, irregular, open, or weak; end of stitching not backstitched; shade of thread on outside not as specified; top stitching too close to edge.
Hook and loop	Any hook and loop out of alignment or not able to be fastened properly; width or length not as specified or not sewn 1/8 to 3/16 inch from the edge
Measurements	Any measurement deviating from nominal dimensions and tolerances specified. Leg inseam and arm inseams uneven by 1 inch or more.
Label	Label missing, incorrect, or illegible
Bar code label/tag	Bar code omitted or not readable by scanner; human-readable interpretation (HRI) omitted or illegible; bar code not visible on folded, packaged, item; bar code causes damage to the item

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Packaging	Item not packaged as specified in the contract or purchase order

5. PACKAGING

5.1 Preservation, packing, and marking. The preservation, packing, and marking shall be as specified in the contract or purchase order.

6. NOTES

6.1 Source of Government documents. Copies of military and Federal documents are available from:

Standardization Documents Order Desk
Bldg. 4D
700 Robbins Avenue
Philadelphia, PA 19111-5094

6.2 Source of non-Government documents

ASTM Test Methods

(Applications for copies should be addressed to American Society For Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

AATCC Test Method

(Applications for copies should be addressed to American Association of Textile Chemists and Colorists (AATCC), P.O. Box 12215, Triangle Park, NC 27709-2215.)

6.3 Suggested sources of supply

A suggested source of supply for the leg bottom mesh is Apex Mills Corporation, 168 Doughty Blvd., Inwood, NY 11696.

A suggested source of supply for the drawcord is W.M Jetty & Son Inc., 68 Salem St., Providence, RI 02907.

A suggested source of supply for the cord lock is I.T.W. Waterbury Buckle, 952 South Main St., Waterbury, CT 06721.

A suggested source of supply for the Retro-reflective Tape is Safety and Security Systems Division/3M, Building 223-3N-01, 3M Center, St. Paul, MN 55144-1000.

6.4 Acceptance criteria. Acceptance criteria shall be as specified in the contract or purchase order.

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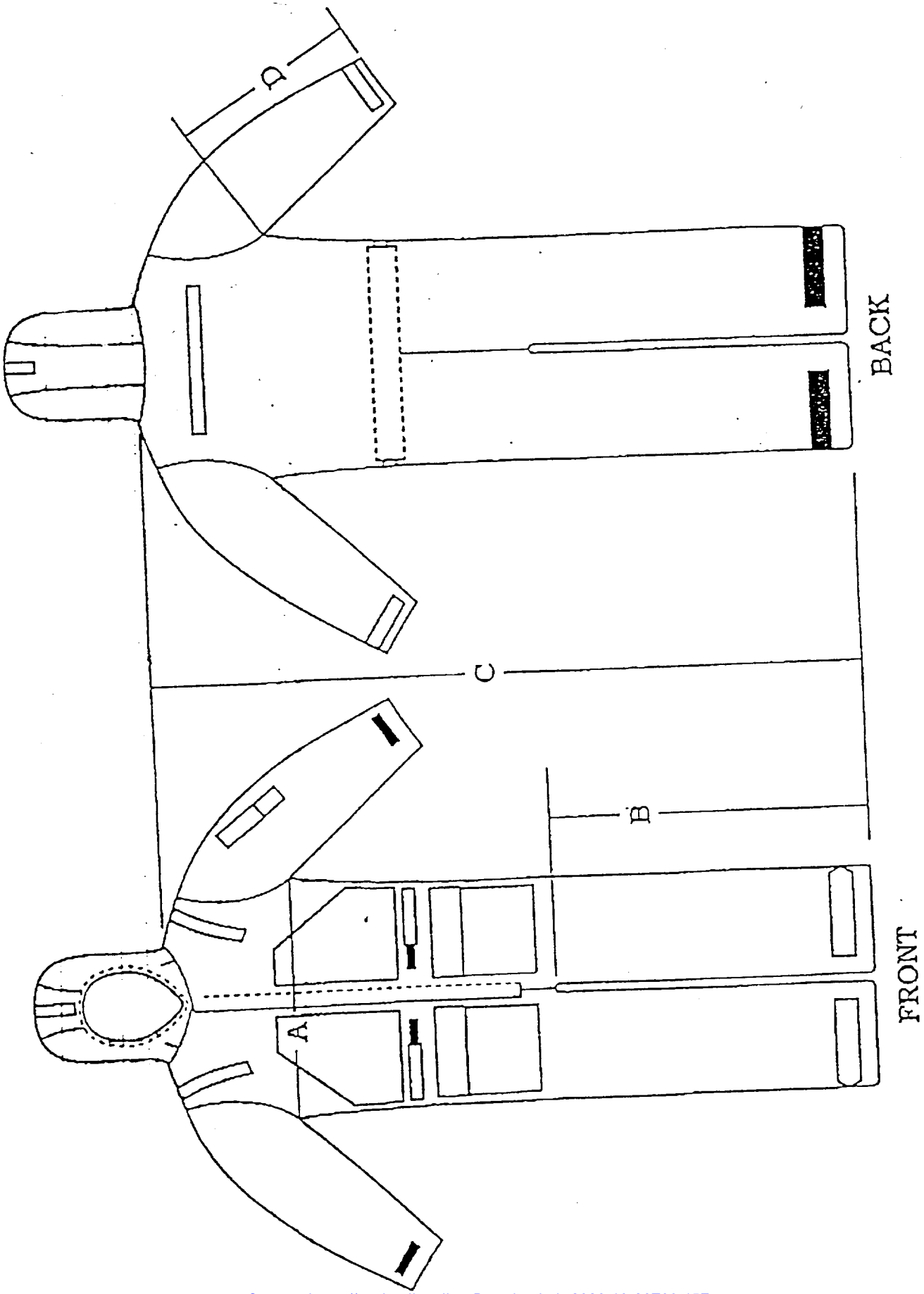


FIGURE 1. Coveralls, Anti-Exposure

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